

Road Event Detector Formulations

road_events documentation

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1 Scope

This document records the math and streaming algorithms implemented by the `road_events` crate. The crate detects road events from vehicle-frame motion signals produced by the simulation and visualization pipeline. The detectors are intentionally small-state and embedded-friendly: they process one sample at a time, use bounded memory, and avoid FFTs, dynamic allocation, and offline window buffering.

The detectors are not part of the EKF state estimator. They consume already estimated or referenced motion channels such as vehicle speed, vehicle pitch, gravity-compensated vertical acceleration, and yaw rate. The current event set is:

- speed bumps,
- sustained uphill and downhill intervals,
- reverse-driving intervals,
- harsh acceleration,
- harsh braking,
- harsh cornering.

2 Signals and Units

Symbol	Unit	Meaning
t_k	s	Sample timestamp.
v_x	m/s	Vehicle-frame forward velocity. Positive is forward; negative is reverse.
$s = v_x $	m/s	Speed magnitude used by threshold logic.
θ	deg	Vehicle pitch angle. Positive pitch is uphill in the current detector convention.
a_z	m/s ²	Gravity-compensated vehicle-frame vertical acceleration.
ω_z	rad/s	Vehicle yaw rate.

The speed-bump detector uses pitch and vertical acceleration. Hill detection uses pitch and speed. Reverse and harsh longitudinal detection use forward velocity. Harsh cornering uses speed and yaw rate.

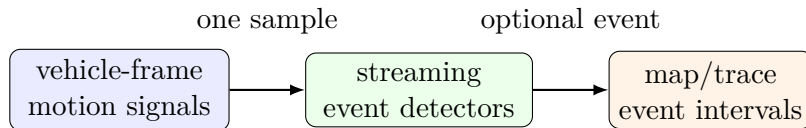


Figure 1: All event detectors run as streaming post-processors over vehicle motion signals.

3 Shared Streaming Primitives

3.1 Timestamp Handling

The implementation clamps per-sample time deltas to keep bad or repeated timestamps from creating unbounded filter updates:

$$\Delta t_k = \text{clamp}(t_k - t_{k-1}, 0, 0.2). \quad (1)$$

For high-pass and derivative paths that require a strictly positive time step, the lower bound is 10^{-3} or 10^{-4} seconds depending on the path.

3.2 First-Order High-Pass Filter

The speed-bump detector removes road grade and slow body motion with the standard discrete first-order high-pass filter:

$$\tau = \frac{1}{2\pi f_c}, \quad (2)$$

$$\alpha_k = \frac{\tau}{\tau + \Delta t_k}, \quad (3)$$

$$y_k = \alpha_k(y_{k-1} + u_k - u_{k-1}). \quad (4)$$

This is applied independently to vehicle pitch and gravity-compensated vertical acceleration.

3.3 Exponential Moving Average

The crate uses an exponential moving average (EMA) for adaptive noise floors and smoothed velocity derivatives:

$$\alpha_k = \frac{\Delta t_k}{\max(\tau, \Delta t_k) + \Delta t_k}, \quad (5)$$

$$\bar{x}_k = (1 - \alpha_k)\bar{x}_{k-1} + \alpha_k x_k. \quad (6)$$

For noise floors, the input is the absolute high-pass value:

$$\sigma_k = \text{EMA}(\sigma_{k-1}, |y_k|, \tau, \Delta t_k). \quad (7)$$

The EMA is a low-memory substitute for a sliding RMS window. It provides slow adaptation to background vibration without storing historical samples.

3.4 Interval Accumulation

Most detectors convert an instantaneous metric m_k into an event interval using enter and exit thresholds:

$$\text{enter if } m_k \geq m_{\text{enter}}, \quad (8)$$

$$\text{stay active while } m_k \geq m_{\text{exit}}, \quad m_{\text{exit}} < m_{\text{enter}}. \quad (9)$$

During an active interval the detector accumulates:

$$T = \sum_k \Delta t_k, \quad (10)$$

$$\bar{m} = \frac{1}{T} \sum_k m_k \Delta t_k, \quad (11)$$

$$m_{\max} = \max_k m_k, \quad (12)$$

$$\bar{s} = \frac{1}{T} \sum_k s_k \Delta t_k. \quad (13)$$

The interval is emitted only if $T \geq T_{\min}$. A refractory time suppresses duplicate events immediately after a valid event.

4 Speed-Bump Detector

4.1 Physical Signature

A road speed bump creates a short vertical impulse sequence as the front axle and rear axle traverse the bump. In gravity-compensated vehicle-frame vertical acceleration, the idealized pattern is an alternating sequence of three extrema:

$$a_{z,\text{hp}}(t_1), \quad a_{z,\text{hp}}(t_2), \quad a_{z,\text{hp}}(t_3), \quad \text{sgn}(a_1) = \text{sgn}(a_3) \neq \text{sgn}(a_2). \quad (14)$$

Pitch motion is used as corroboration rather than as the primary trigger, because pitch angle alone can also respond to braking, turning on uneven roads, or grade changes.

4.2 Filtered Signals

The detector computes:

$$\theta_{\text{hp},k} = \text{HPF}(\theta_k; f_c = 0.45 \text{ Hz}), \quad (15)$$

$$a_{\text{hp},k} = \text{HPF}(a_{z,k}; f_c = 0.70 \text{ Hz}). \quad (16)$$

Adaptive noise floors are:

$$\sigma_{\theta,k} = \text{EMA}(|\theta_{\text{hp},k}|; \tau = 6 \text{ s}), \quad (17)$$

$$\sigma_{a,k} = \text{EMA}(|a_{\text{hp},k}|; \tau = 6 \text{ s}). \quad (18)$$

The peak thresholds are:

$$A_{\text{thr}} = \max(3.5 \sigma_a, 1.5 \text{ m/s}^2), \quad (19)$$

$$\Theta_{\text{thr}} = \max(3.0 \sigma_{\theta}, 0.25^\circ). \quad (20)$$

4.3 Extremum Extraction

The vertical acceleration slope is

$$\dot{a}_{\text{hp},k} = \frac{a_{\text{hp},k} - a_{\text{hp},k-1}}{\Delta t_k}. \quad (21)$$

A local maximum is detected when $\dot{a}_{k-1} > 0$ and $\dot{a}_k \leq 0$; a local minimum is detected when $\dot{a}_{k-1} < 0$ and $\dot{a}_k \geq 0$. An extremum is kept only when its absolute vertical acceleration exceeds the physical floor 1.5 m/s^2 . The detector stores only the latest four extrema, so memory use remains constant.

For each extremum interval, the detector also stores the maximum pitch high-pass magnitude since the previous extremum:

$$\Theta_i = \max_{t \in (t_{i-1}, t_i]} |\theta_{\text{hp}}(t)|. \quad (22)$$

4.4 Candidate Timing

Each latest-three-extrema group is scored only if the polarities alternate. Let

$$D = t_3 - t_1, \quad \bar{s} = \frac{s_1 + s_2 + s_3}{3}. \quad (23)$$

The candidate must pass both absolute and speed-adaptive duration bounds:

$$D_{\min} = \max \left(0.18, 0.35 \frac{L_{\min}}{\bar{s}} \right), \quad (24)$$

$$D_{\max} = \min \left(1.8, 2.5 \frac{L_{\max}}{\bar{s}} \right), \quad (25)$$

where $L_{\min} = 1.8 \text{ m}$ and $L_{\max} = 3.6 \text{ m}$ are plausible front/rear axle spacing bounds. The speed must also satisfy $\bar{s} \geq 1.5 \text{ m/s}$.

The detector tracks the total time within the pattern for which $|a_{\text{hp}}| \geq 1.5 \text{ m/s}^2$. It requires:

$$T_{\text{active}} \geq 0.25 \text{ s}, \quad (26)$$

$$\frac{T_{\text{active}}}{D} \geq 0.25. \quad (27)$$

4.5 Score and Confidence

For extrema magnitudes $A_i = |a_i|$, pitch peaks Θ_i , and candidate duration D , the component scores are:

$$A_{\max} = \max_i A_i, \quad (28)$$

$$\Theta_{\max} = \max_i \Theta_i, \quad (29)$$

$$S_a = \text{clamp} \left(\frac{A_{\max}}{A_{\text{thr}}} - 1, 0, 1 \right), \quad (30)$$

$$S_\theta = \text{clamp} \left(\frac{\Theta_{\max}}{\Theta_{\text{thr}}} - 1, 0, 1 \right), \quad (31)$$

$$S_b = \frac{\min(A_1, A_3)}{\max(A_1, A_3, 10^{-3})}. \quad (32)$$

The spacing score favors the center of the admissible duration interval:

$$D_c = \frac{D_{\min} + D_{\max}}{2}, \quad W = \frac{\max(D_{\max} - D_{\min}, 10^{-3})}{2}, \quad (33)$$

$$S_D = \text{clamp} \left(1 - \frac{|D - D_c|}{W}, 0, 1 \right). \quad (34)$$

The final pattern score is:

$$S = \text{clamp}((0.45S_a + 0.25S_D + 0.15S_b + 0.15S_\theta)S_\theta, 0, 1). \quad (35)$$

The pitch score multiplies the weighted score, so vertical acceleration alone does not emit an event.

An event is emitted when $S \geq 0.12$ and at least four seconds have elapsed since the last bump event. The UI-facing confidence is not a calibrated probability; it is normalized only after the trigger threshold is cleared:

$$c_{\text{display}} = 0.90 + 0.08 \text{ clamp}\left(\frac{S - 0.12}{1 - 0.12}, 0, 1\right). \quad (36)$$

5 Hill Detector

The hill detector identifies sustained grade from vehicle pitch. It is a piecewise-constant interval detector with sign:

$$\text{kind}(\theta) = \begin{cases} \text{uphill}, & \theta \geq 4^\circ, \\ \text{downhill}, & \theta \leq -4^\circ, \\ \emptyset, & \text{otherwise.} \end{cases} \quad (37)$$

An interval is emitted when the same signed condition remains active for at least three seconds. During the interval, the detector reports:

$$T = t_{\text{end}} - t_{\text{start}}, \quad (38)$$

$$\bar{\theta} = \frac{1}{T} \int_{t_{\text{start}}}^{t_{\text{end}}} \theta(t) dt, \quad (39)$$

$$\theta_{\text{max}} = \max_t |\theta(t)|, \quad (40)$$

$$\bar{s} = \frac{1}{T} \int_{t_{\text{start}}}^{t_{\text{end}}} \max(s(t), 0) dt. \quad (41)$$

The implementation emits the event when pitch exits the threshold band or when the detector is explicitly finished at the end of a replay.

6 Reverse Detector

Reverse detection uses vehicle-frame forward velocity directly. The detector starts a candidate when:

$$v_x < -0.5 \text{ m/s}. \quad (42)$$

The candidate is confirmed only after this condition persists for 0.5 s. Once confirmed, the interval remains active while:

$$v_x < -0.2 \text{ m/s}. \quad (43)$$

This hysteresis prevents repeated short events around zero speed. The event is closed after 0.5 s above the exit threshold and emitted if its duration is at least one second.

Reported reverse speed uses positive magnitude:

$$s_{\text{rev}} = \max(-v_x, 0). \quad (44)$$

Mean and peak reverse speeds are accumulated over the confirmed interval.

7 Harsh Acceleration and Braking

7.1 Velocity-Derivative Metric

Harsh longitudinal events are based on changes in forward velocity rather than raw accelerometer samples. This makes the detector less sensitive to road vibration and IMU mounting artifacts. The raw derivative is:

$$a_{x,\text{raw},k} = \text{clamp}\left(\frac{v_{x,k} - v_{x,k-1}}{\Delta t_k}, -15, 15\right) \text{ m/s}^2. \quad (45)$$

The metric used by the interval detector is an EMA-smoothed derivative:

$$\hat{a}_{x,k} = \text{EMA}(\hat{a}_{x,k-1}, a_{x,\text{raw},k}, \tau = 0.6 \text{ s}, \Delta t_k). \quad (46)$$

The speed gate uses:

$$s_k = \max(|v_{x,k}|, |v_{x,k-1}|). \quad (47)$$

7.2 Harsh Acceleration

If $s_k < 1 \text{ m/s}$, the metric is forced to zero. Otherwise:

$$m_{\text{accel},k} = \max(\hat{a}_{x,k}, 0). \quad (48)$$

The interval thresholds are:

$$m_{\text{enter}} = 2.5 \text{ m/s}^2, \quad (49)$$

$$m_{\text{exit}} = 2.0 \text{ m/s}^2, \quad (50)$$

$$T_{\text{min}} = 0.4 \text{ s}, \quad (51)$$

$$T_{\text{refractory}} = 2.0 \text{ s}. \quad (52)$$

The emitted event reports duration, mean and peak acceleration metric, mean and peak speed, and net velocity change:

$$\Delta v_x = v_{x,\text{end}} - v_{x,\text{start}}. \quad (53)$$

7.3 Harsh Braking

Braking uses the same smoothed derivative but flips the sign:

$$m_{\text{brake},k} = \max(-\hat{a}_{x,k}, 0). \quad (54)$$

The default thresholds are:

$$m_{\text{enter}} = 3.0 \text{ m/s}^2, \quad (55)$$

$$m_{\text{exit}} = 2.4 \text{ m/s}^2, \quad (56)$$

$$T_{\text{min}} = 0.4 \text{ s}, \quad (57)$$

$$T_{\text{refractory}} = 2.0 \text{ s}. \quad (58)$$

The reported Δv_x is signed, so braking events normally have $\Delta v_x < 0$.

8 Harsh Cornering

Harsh cornering uses the steady-turn kinematic relation:

$$a_y \approx v\omega_z. \quad (59)$$

The detector metric is:

$$m_{\text{corner},k} = \begin{cases} |s_k\omega_{z,k}|, & s_k \geq 3 \text{ m/s}, \\ 0, & s_k < 3 \text{ m/s}. \end{cases} \quad (60)$$

The default interval thresholds are:

$$m_{\text{enter}} = 3.0 \text{ m/s}^2, \quad (61)$$

$$m_{\text{exit}} = 2.4 \text{ m/s}^2, \quad (62)$$

$$T_{\text{min}} = 0.5 \text{ s}, \quad (63)$$

$$T_{\text{refractory}} = 2.0 \text{ s}. \quad (64)$$

The steady-turn approximation is deliberately simple. It is robust for embedded event detection because it only needs speed and yaw rate, and it does not depend on lateral accelerometer calibration or road bank compensation.

9 Default Configuration Summary

Detector	Default	Purpose
Speed bump pitch HPF	0.45 Hz	Reject grade and slow pitch.
Speed bump vertical accel HPF	0.70 Hz	Reject slow body motion.
Speed bump noise EMA	6.0 s	Adaptive vibration floor.
Speed bump minimum speed	1.5 m/s	Avoid static/creep false positives.
Speed bump duration	0.18–1.8 s	Absolute event timing.
Speed bump wheelbase bounds	1.8–3.6 m	Speed-adaptive timing.
Hill pitch threshold	4°	Sustained uphill/downhill grade.
Hill minimum duration	3.0 s	Suppress short pitch transients.
Reverse enter/exit	−0.5/ − 0.2 m/s	Hysteresis around zero speed.
Reverse minimum duration	1.0 s	Suppress brief backing corrections.
Harsh accel enter/exit	2.5/2.0 m/s ²	Longitudinal acceleration hysteresis.
Harsh brake enter/exit	3.0/2.4 m/s ²	Longitudinal deceleration hysteresis.
Harsh corner enter/exit	3.0/2.4 m/s ²	Lateral acceleration hysteresis.

10 Visualizer Interpretation

The map markers and hover mini-plots are visualization aids. They do not feed back into the detectors. For each event, the visualizer draws a short local window around the detected interval,

currently with five seconds of padding on both sides when data is available:

$$[t_{\text{start}} - 5, t_{\text{end}} + 5]. \tag{65}$$

The mini-plot shows the signal that caused the detector to trigger, for example vertical acceleration and pitch for speed bumps, longitudinal acceleration for harsh acceleration/braking, and yaw-rate-derived lateral acceleration for harsh cornering.

11 Design Rationale

The event crate favors deterministic low-memory logic over more expressive offline methods. Windowed FFTs, matched filters, and machine-learning classifiers can be useful for richer offline analysis, but they introduce larger buffers, more tuning surface, and more deployment complexity. The current detectors instead use physically interpretable metrics:

- speed bumps: alternating high-pass vertical acceleration extrema corroborated by pitch motion,
- hills: sustained pitch sign and magnitude,
- reverse: signed vehicle-frame forward velocity with hysteresis,
- harsh acceleration/braking: smoothed velocity derivative,
- harsh cornering: steady-turn lateral acceleration $v\omega_z$.

This keeps the algorithms easy to port to small embedded targets and easy to debug from the visualizer.